SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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1) Intelligent agents in Smart home automation

Introduction:-

An intelligent agent observes its environment throu
gh sensors and takes actions to achieve goals

1.) Simple Reflex Agent

Acts based on current inputs. Example Switches on lights when nobody is in the room.

2.) Model-Based Agent

Uses current inputs and past information. Example: Switches off lights when nobody is in the room

3.) Goal-Based Agent

Selects actions to achieve a specific goal. Example:- Adjusts room temperature to the desired level.

4.) Utility-Based Agent

Chooses the action that provides that greatest benefit. Example:- Maintains comfort while saving electricity.

Best Approach:-

A hybrid approach combining all four agents is best because it provides comfort, security, and energy efficiency.

Conclusion:-

Intellignet agents make smart homes safer, smarter and more efficient.

Q:) UCS and IDS in Maze solving

Introduction:-

Uninformed search algorithms explore paths to find the exit without additional knowledge about the goal's location.

1.) Uniform cost Search (UCS)

UCS expands the path with the lowest total cost using a priority queue.

Working:-

1.) Start from the initial position
2.) Select the lowest-cost path
3.) Check for the exit.
4.) Repeat until the next is found.

Advantages:- Finds the least-cost is found.

Disadvantages:- Requires more memory and time

2.) Iterative Deepening Search (IDS)

IDS repeatedly searches the maze by increasing the depth limit until the exit is found.

Advantages:- Uses less memory and finds the shallowest solution when all step costs are equal.

Disadvantages:- Repeats searches and my waste time.

Complexity:-

- UCS:- Exponential time & space in the general case
- IDS:- Time (O(b^d)), space (O(bd)).

Here, (b) is the branching factor and (d) is the solution depth.

Intelligent Agents:-

- Simple Reflex:- Acts on current inputs.
- Model-Based:- Remembers visited positions.
- Goal-Based:- Plans to reach the exit.
- Utility-Based:- chooses paths based on cost & energy

Best Approach:-

A model-based, goal-based agent is suitable. UCS is best for different movement costs, while IDS is suitable for equal movement costs and limited memory.

Conclusion:-

UCS finds the least-cost path, whereases IDS saves memory. The appropriate algorithm helps the robot navigate the maze efficiently.